Audio Video Bridging: Evolution of Multimedia Streaming Technology

The Institute of Electrical and Electronics Engineers (IEEE) has developed a set of technical standards that allow time-synchronised low latency streaming services through IEEE 802 networks. This is commonly known as audio video bridging. Read on to learn more about the concept.

Initially, when our project was instituted, I was oblivious of the potential of the audio video bridging (AVB) standard and its applications, as I was mainly involved in the protocol implementation to get the desired outcome. It was only when I started monitoring AVB products and the AVB demonstrations held by companies in this domain that I learned about its significance.

Let us start with a brief introduction to Ethernet AVB, which has evolved from the existing Ethernet standard. This technology ensures tight control over the quality of service required for streaming real-time high-quality audio and video over wired Ethernet. Before the existence of AVB standards, the audio and video set-up involved a large number of cables intertwined with each other. Now, AVB technology provides a high quality audio and video experience using structured wiring without any complicated set-ups, while coexisting with the other non-media network traffic. AVB technology involves fundamental changes to the Ethernet standards to support the audio-video media distribution, by furnishing high precision network synchronisation, dynamic bandwidth reservation to ensure packet delivery within stipulated latency and traffic shaping to minimise bandwidth needs. It also eliminates media bursts even when passing through multiple network hops, and provides admission control to ensure that non-AVB devices don't degrade the performance of the AVB network and devices. The AVB ecosystem is evolving and creating its own niche in the automotive infotainment, consumer electronics, broadcast system design and other domains.

Before the AVB standard came into existence, there was no standard way to synchronise the audio-video devices, to reserve the bandwidth and resources to stream real-time AV data between these devices, and to discover and control them. To understand the functionality of AVB, assume you have listener devices - one speaker in the kitchen, one in the garage and one in your bedroom. You have a talker device, like a mobile, connected with the three speakers in an AVB network, and you wish to stream AV data to these listener devices in real-time. You also want to download data from the Internet, simultaneously, on your mobile device. The real-time streaming will not be affected even when non-AVB traffic faces traffic congestion, and the AV data will be played at the same instant by all the listeners.

AVB is the name given to a set of protocol standards developed by the IEEE Audio Video Bridging Task Group. To standardise and improve the interoperability of AVB devices and networks, a consortium of automotive and consumer electronics companies, called the AVnu Alliance, was formed in August 2009. It constantly strives to establish and certify AVB products which